

CanonicalForms

Abstract

We formulate, in the language of set theory, the concepts of invariant and canonical form for a binary relation $\sim$ on a set X , and study them mainly when $\sim$ is an equivalence relation. A map $f: X \rightarrow Y$ is called an invariant of $\sim$ if $x \sim y \implies f(x) = f(y)$, and a complete invariant if equality of images also implies $x \sim y$. A canonical form is an invariant $C: X \rightarrow X$ satisfying $x \sim C(x)$ for all $x \in X$; its image $C(X)$ is called the skeleton of $\sim$, and, when $\sim$ is an equivalence relation, also the complete system of representatives of $\sim$. We characterise canonical forms via retraction–section pairs, showing that C is a canonical form if and only if there exists a section s of the quotient map $q: X \rightarrow X/\sim$ such that $C = s \circ q$. An analogous characterisation holds for surjective complete invariants. We prove that every complete invariant $I: X \rightarrow Y$ factors uniquely through the surjective complete invariant $I': X \rightarrow I(X)$. For a section s' of I' , we call $s' \circ I'$ the canonical form of $\sim$ determined by I and s' , and show that it is a canonical form of $\sim$. Assuming the axiom of choice, we show that I' has a section, so that such a canonical form exists for every complete invariant I . Propositions that assume the axiom of choice are marked with ^{AC}.

Definition 1 Let X, Y be sets. Let $f: X \rightarrow Y$ be a map and let $\sim$ be a binary relation on X . We say that f is an **invariant** of $\sim$ if the following condition holds.

$$\forall x, y \in X, (x \sim y \implies f(x) = f(y))$$

We further say that f is a **complete invariant** of $\sim$ if the following condition holds.

$$\forall x, y \in X, (x \sim y \iff f(x) = f(y))$$

□

Example 1 Let X be a set. Let $\sim$ be an equivalence relation on X and let $q: X \rightarrow X/\sim$ be the quotient map. Then q is a complete invariant of $\sim$. □

Definition 2 Let X be a set. Let $C: X \rightarrow X$ be an invariant of a binary relation $\sim$ on X . We say that C is a **canonical form** of $\sim$ if the following condition holds.

$$\forall x \in X, x \sim C(x)$$

We call the image $C(X)$ of X under C the **skeleton** of $\sim$. When $\sim$ is an equivalence relation, we also call $C(X)$ the **complete system of representatives** of $\sim$. $\square$

Proposition 1 Let X be a set. Let $C: X \rightarrow X$ be a canonical form of a binary relation $\sim$ on X . Then $C = C \circ C$. $\square$

Proof Let $x \in X$. Since $x \sim C(x)$, we have $C(x) = C(C(x))$. Therefore $C = C \circ C$. $\blacksquare$

Definition 3 Let X, Y be sets and let $r: X \rightarrow Y$ and $s: Y \rightarrow X$ be maps. We call the pair (r, s) a **retraction-section pair** if the following condition holds. We call r the **retraction** of s , and s the **section** of r .

$$r \circ s = \text{id}_Y$$

$\square$

Lemma 1 Let X be a set and let $q: X \rightarrow X/\sim$ be the quotient map of an equivalence relation $\sim$ on X . For a map $C: X \rightarrow X$, the following conditions (1) and (2) are equivalent.

- (1) C is a canonical form of $\sim$.
- (2) There exists a section $s: X/\sim \rightarrow X$ of $q: X \rightarrow X/\sim$ such that $C = s \circ q$.

$\square$

Proof (1) $\Rightarrow$ (2): Assume C is a canonical form of $\sim$. For each $y \in X/\sim$, since q is surjective there exists $x \in X$ such that $y = q(x)$. Set $s(y) = C(x)$. For any $x' \in X$ with $q(x) = q(x')$, we have $x \sim x'$, so $C(x) = C(x')$. Thus $s: X/\sim \rightarrow X$ is well-defined. For any $x \in X$, $C(x) = s(q(x))$, so $C = s \circ q$. For any $x \in X$, $(q \circ s)(q(x)) = q(s(q(x))) = q(C(x))$, and since $x \sim C(x)$ we have $q(C(x)) = q(x)$. Therefore $q \circ s = \text{id}_{X/\sim}$.

(2) $\Rightarrow$ (1): Assume there exists a section $s: X/\sim \rightarrow X$ of $q: X \rightarrow X/\sim$ such that $C = s \circ q$. For $x, x' \in X$, if $x \sim x'$, then $C(x) = (s \circ q)(x) = s(q(x)) = s(q(x')) = (s \circ q)(x') = C(x')$, so C is an invariant of $\sim$. For any $x \in X$, $q(x) = \text{id}_{X/\sim}(q(x)) = (q \circ s)(q(x)) = q((s \circ q)(x)) = q(C(x))$, so $x \sim C(x)$. Therefore C is a canonical form of $\sim$. $\blacksquare$

Proposition 2 Let X, Y be sets and let $I: X \rightarrow Y$ be a surjective complete invariant of an equivalence relation $\sim$ on X . For a map $C: X \rightarrow X$, the following conditions (1) and (2) are equivalent.

- (1) C is a canonical form of $\sim$.
- (2) There exists a section $s': Y \rightarrow X$ of $I: X \rightarrow Y$ such that $C = s' \circ I$. $\square$

Proof By Lemma 1, it suffices to show that the existence of a section $s: X/\sim \rightarrow X$ of $q: X \rightarrow X/\sim$ with $C = s \circ q$ is equivalent to (2). Define $\bar{I}: X/\sim \rightarrow Y$ by $\bar{I}(q(x)) = I(x)$ for all $x \in X$. When $x \sim x'$, since I is an invariant we have $I(x) = I(x')$, so $\bar{I}$ is well-defined. For any $x, x' \in X$, if $I(x) = I(x')$ then, since I is a complete invariant, $x \sim x'$, hence $q(x) = q(x')$, so $\bar{I}$ is injective. For any $y \in Y$, since $I: X \rightarrow Y$ is surjective there exists $x \in X$ with $I(x) = y$, giving $\bar{I}(q(x)) = y$. Therefore $\bar{I}$ is bijective and hence invertible.

Suppose there exists a section s of q with $C = s \circ q$. Set $s' = s \circ \bar{I}^{-1}$. Since $q = \bar{I}^{-1} \circ I$, we have $C = s \circ \bar{I}^{-1} \circ I = s' \circ I$. Moreover, $I \circ s' = (\bar{I} \circ q) \circ (s \circ \bar{I}^{-1}) = \bar{I} \circ (q \circ s) \circ \bar{I}^{-1} = \bar{I} \circ \text{id}_{X/\sim} \circ \bar{I}^{-1} = \text{id}_Y$, so s' is a section of I .

Conversely, suppose (2) holds. Set $s = s' \circ \bar{I}$. Then $s \circ q = s' \circ \bar{I} \circ q = s' \circ I = C$. Since $I \circ s' = \text{id}_Y$, we have $I \circ s' \circ I = I$, and therefore $I \circ C = I$. Since $I = \bar{I} \circ q$, it follows that $\bar{I} \circ q \circ C = \bar{I} \circ q$, and by injectivity of $\bar{I}$ we obtain $q \circ C = q$. For any $y \in Y$ there exists $x \in X$ such that $y = q(x)$. Then

$$(q \circ s)(y) = (q \circ s' \circ \bar{I})(y) = (q \circ s' \circ \bar{I} \circ q)(x) = (q \circ s' \circ I)(x) = (q \circ C)(x) = q(x) = y,$$

so $q \circ s = \text{id}_{X/\sim}$. ■

Proposition 3 Let X, Y be sets and let $I: X \rightarrow Y$ be a complete invariant of an equivalence relation $\sim$ on X . The unique map $I': X \rightarrow I(X)$ satisfying $I = i \circ I'$ with respect to the inclusion $i: I(X) \rightarrow Y$ is surjective and is a complete invariant of $\sim$. □

Proof Define $I': X \rightarrow I(X)$ by $I'(x) = I(x)$ for all $x \in X$. Then $i(I'(x)) = I(x)$ for all $x \in X$, so $I = i \circ I'$; this shows that such a map exists. Since $I'(X) = I(X)$, I' is surjective. For any $x, x' \in X$, $x \sim x' \iff I(x) = I(x') \iff I'(x) = I'(x')$, so I' is a complete invariant of $\sim$. If $I'': X \rightarrow I(X)$ also satisfies $I = i \circ I''$, then $i(I'(x)) = I(x) = i(I''(x))$ for all $x \in X$, and since i is injective, $I'(x) = I''(x)$. Therefore $I'' = I'$, so I' is unique. ■

Definition 4 Let X, Y be sets and let $I: X \rightarrow Y$ be a complete invariant of an equivalence relation $\sim$ on X . Let $I': X \rightarrow I(X)$ be the unique map satisfying $I = i \circ I'$ with respect to the inclusion $i: I(X) \rightarrow Y$, and let $s': I(X) \rightarrow X$ be a section of I' . We call the map $s' \circ I': X \rightarrow X$ the **canonical form of $\sim$ determined by I and s'** . □

Proposition 4 In the setting of Definition 4, the canonical form $s' \circ I'$ of $\sim$ determined by I and s' is a canonical form of $\sim$. □

Proof By Proposition 3, $I': X \rightarrow I(X)$ is a surjective complete invariant of $\sim$. Apply Proposition 2 to I' , with Y replaced by $I(X)$, and to $C = s' \circ I'$. Since s' is a section of I' with $C = s' \circ I'$, condition (2) holds, and hence $s' \circ I'$ is a canonical form of $\sim$. ■

Lemma 2^{AC} Let X, Y be sets and let $f: X \rightarrow Y$ be a surjection. Then there exists a map $g: Y \rightarrow X$ such that $f \circ g = \text{id}_Y$. □

Proof Since $f: X \rightarrow Y$ is surjective, $f^{-1}(y) \neq \emptyset$ for every $y \in Y$. Therefore, by the axiom of choice, $\prod_{y \in Y} f^{-1}(y) \neq \emptyset$, so there exists $g \in \prod_{y \in Y} f^{-1}(y)$. This map $g: Y \rightarrow X$ satisfies $f \circ g = \text{id}_Y$. ■

Proposition 5^{AC} Let X, Y be sets and let $I: X \rightarrow Y$ be a complete invariant of an equivalence relation $\sim$ on X . Let $I': X \rightarrow I(X)$ be the unique map satisfying $I = i \circ I'$ with respect to the inclusion $i: I(X) \rightarrow Y$. Then there exists a section $s': I(X) \rightarrow X$ of I' , and hence there exists a canonical form of $\sim$ determined by I and s' . □

Proof By Proposition 3, I' is surjective, so by Lemma 2 there exists a map $s': I(X) \rightarrow X$ satisfying $I' \circ s' = \text{id}_{I(X)}$, that is, a section of I' . By Proposition 4, the canonical form $s' \circ I'$ of $\sim$ determined by I and s' is a canonical form of $\sim$. ■